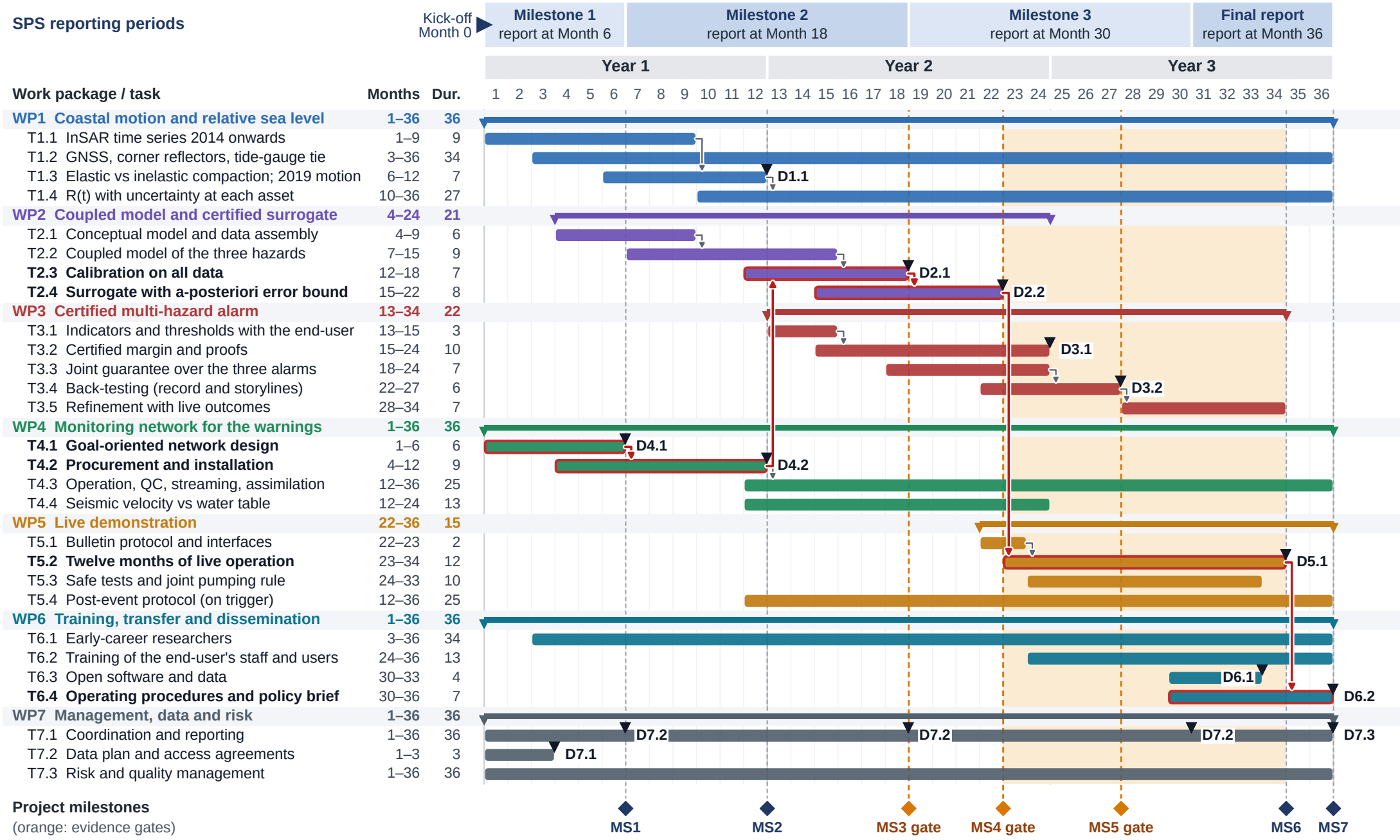
CERTAIN-COAST — Project schedule (36 months)

Certified early warning to protect water, port and energy infrastructure on a sinking Adriatic coast. Months count from the SPS kick-off meeting (Month 0).



Work package / task	Months	Dur.
WP1 Coastal motion and relative sea level	**1–36**	**36**
T1.1 InSAR time series 2014 onwards	1–9	9
T1.2 GNSS, corner reflectors, tide-gauge tie	3–36	34
T1.3 Elastic vs inelastic compaction; 2019 motion	6–12	7
T1.4 R(t) with uncertainty at each asset	10–36	27
WP2 Coupled model and certified surrogate	**4–24**	**21**
T2.1 Conceptual model and data assembly	4–9	6
T2.2 Coupled model of the three hazards	7–15	9
T2.3 Calibration on all data	12–18	7
T2.4 Surrogate with a-posteriori error bound	15–22	8
WP3 Certified multi-hazard alarm	**13–34**	**22**
T3.1 Indicators and thresholds with the end-user	13–15	3
T3.2 Certified margin and proofs	15–24	10
T3.3 Joint guarantee over the three alarms	18–24	7
T3.4 Back-testing (record and storylines)	22–27	6
T3.5 Refinement with live outcomes	28–34	7
WP4 Monitoring network for the warnings	**1–36**	**36**
T4.1 Goal-oriented network design	1–6	6
T4.2 Procurement and installation	4–12	9
T4.3 Operation, QC, streaming, assimilation	12–36	25
T4.4 Seismic velocity vs water table	12–24	13
WP5 Live demonstration	**22–36**	**15**
T5.1 Bulletin protocol and interfaces	22–23	2
T5.2 Twelve months of live operation	23–34	12
T5.3 Safe tests and joint pumping rule	24–33	10
T5.4 Post-event protocol (on trigger)	12–36	25
WP6 Training, transfer and dissemination	**1–36**	**36**
T6.1 Early-career researchers	3–36	34
T6.2 Training of the end-user's staff and users	24–36	13
T6.3 Open software and data	30–33	4
T6.4 Operating procedures and policy brief	30–36	7
WP7 Management, data and risk	**1–36**	**36**
T7.1 Coordination and reporting	1–36	36
T7.2 Data plan and access agreements	1–3	3
T7.3 Risk and quality management	1–36	36

Project milestones
(orange: evidence gates)

Task (colour = work package) · Critical path · Needs (red: on the critical path) · Deliverable · Milestone / gate · Live operation, Months 23–34

A task "12–18" runs from the start of Month 12 to the end of Month 18. Every dependency, with what it passes on, is listed on the next page; deliverables by SPS milestone and the buffers follow.

CERTAIN-COAST — Steps, durations and dependencies

7 work packages, 28 tasks, 36 months. "Needs" lists the tasks each step depends on and what they pass on; a step that needs one still running starts on its preliminary results. Tasks on the critical path, whose chain decides the end date, are in bold.

Task	Step	Months	Dur.	Needs (what it passes on)	Critical
WP1	**Coastal motion and relative sea level**	1–36	36		
T1.1	Sentinel-1 InSAR time series (2014 onwards) over the three asset areas	1–9	9	Starts at kick-off (open Sentinel-1 archive)	
T1.2	GNSS station(s) and corner reflectors at the assets; tie to the tide gauge	3–36	34	T7.2 (site access)	
T1.3	Elastic and inelastic compaction separated; pumping limit without inelastic compaction; 2019 coseismic and postseismic motion	6–12	7	T1.1 (InSAR time series)	
T1.4	R(t) series with uncertainty at each asset, updated through the demonstration	10–36	27	T1.2 (GNSS tie to the tide gauge); T1.3 (compaction separated)	
WP2	**Coupled model and certified surrogate**	4–24	21		
T2.1	Conceptual model and data assembly: geology, wells, port and landfall geotechnical data	4–9	6	T7.2 (Survey data)	
T2.2	Coupled model: variable-density flow, salt memory, compaction, FS(x,t) = CRR/CSR(z_w), coastal water level on R(t)	7–15	9	T2.1 (conceptual model and data)	
T2.3	**Calibration on heads, conductivity, chloride, InSAR and seismic velocity**	12–18	7	T2.2 (model); T4.2 (data streaming); T1.3 (land motion)	Yes
T2.4	**Neural or kernel operator surrogate with a-posteriori error bound**	15–22	8	T2.3 (calibrated parameters; preliminary ones first)	Yes
WP3	**Certified multi-hazard alarm**	13–34	22		
T3.1	Hazard indicators and thresholds agreed with the end-user (chloride limit, FS < 1, design water level)	13–15	3	T2.2 (model outputs that define the indicators)	
T3.2	Margin construction and proofs: error bound + online conformal risk control + extreme-aware term	15–24	10	T3.1 (thresholds); T2.4 (surrogate error bound)	
T3.3	Joint guarantee over the three alarms; tolerances α_k allocated by consequence	18–24	7	T3.2 (margin of each alarm)	
T3.4	Back-testing on the historical record and on storylines beyond it	22–27	6	T3.3 (joint guarantee); T2.4 (storylines from the surrogate)	
T3.5	Alarm refinement with the outcomes of the live operation	28–34	7	T3.4 (back-testing, MS5); T5.2 (live outcomes)	
WP4	**Monitoring network for the warnings**	1–36	36		
T4.1	**Goal-oriented design across sensor types: piezometers with conductivity, chloride sampling, GNSS, seismic noise, nodes, fibre**	1–6	6	Starts at kick-off (existing and remote-sensing data)	Yes
T4.2	**Procurement and installation**	4–12	9	T4.1 (design; sensors sure to be needed ordered in Month 4)	Yes
T4.3	Operation, quality control, data streaming and assimilation	12–36	25	T4.2 (installed network)	
T4.4	Seismic velocity calibrated against the water table at the port or landfall	12–24	13	T4.2 (installed seismic sensors and piezometers)	
WP5	**Live demonstration**	22–36	15		
T5.1	Bulletin protocol and decision interfaces with the end-user	22–23	2	T3.1 (thresholds); T3.2 (alarm margins)	
T5.2	**Twelve months of live operation, bulletins logged and outcomes verified**	23–34	12	T2.4 (surrogate passes gate MS4); T5.1 (bulletin protocol)	Yes
T5.3	Safe pumping tests planned to sharpen the certified window; pumping rule evaluated against the three hazards together and prepared for the authorities that issue water-use permits	24–33	10	T2.4 (surrogate); T5.2 (live operation)	
T5.4	Post-event protocol after strong earthquakes or surges (on trigger)	12–36	25	T4.2 (network installed)	
WP6	**Training, transfer and dissemination**	1–36	36		
T6.1	Early-career researchers: exchanges, schools and supervised work in WP1–WP5	3–36	34	Starts with recruitment; supervised work in WP1–WP5	
T6.2	Training of the end-user's staff and of the intended users	24–36	13	T5.1 (bulletin protocol); T5.2 (live operation)	
T6.3	Open software and data release with documentation	30–33	4	T3.4 (back-tested algorithms); T2.4 (surrogate)	
T6.4	**Operating procedures and policy brief**	30–36	7	T5.2 (live operation)	Yes
WP7	**Management, data and risk**	1–36	36		
T7.1	Coordination, steering committee, reporting	1–36	36	Runs throughout	
T7.2	Data management, drilling permits and access agreements for existing wells	1–3	3	Starts at kick-off	
T7.3	Risk and quality management	1–36	36	Runs throughout	

Milestones and deliverables by SPS reporting milestone

The SPS reporting milestones are those of the application form and of the Info tab of the budget workbook. Each report gives the project milestones reached in its period, with the result of their pass tests, and the deliverables due in it.

SPS milestone	Project milestones reached, with their pass tests	Deliverables due
Kick-off, Month 0	Project start; steering committee constituted; access agreements started (T7.2)	—
Milestone 1, Month 6 First report on Months 1–6	**MS1** (M6) Network design agreed. Pass: D4.1 approved by the steering committee	D7.1 (M3) Data management plan and access agreements D4.1 (M6) Network design report D7.2 (M6) Progress report to SPS
Milestone 2, Month 18 Second report on Months 7–18	**MS2** (M12) Network streaming; coastal motion known. Pass: D4.2 (≥ 90% data availability over the first month) and D1.1 delivered **MS3** (M18) Coupled model calibrated (evidence gate). Pass: D2.1 shows held-out observations reproduced within their stated uncertainty	D1.1 (M12) Coastal motion atlas and R(t) series D4.2 (M12) Network installed and streaming D2.1 (M18) Calibrated coupled model D7.2 (M18) Progress report to SPS
Milestone 3, Month 30 Third report on Months 19–30	**MS4** (M22) Go/no-go for live operation (evidence gate). Pass: D2.2 surrogate error bound met on the test set; alarm prototype passes a dry run **MS5** (M27) Miss-rate targets confirmed in back-testing (evidence gate). Pass: D3.2 shows each miss rate $\leq \alpha_k$ with its confidence interval reported	D2.2 (M22) Certified surrogate D3.1 (M24) Certified alarm algorithms and proofs D3.2 (M27) Back-testing report D7.2 (M30) Progress report to SPS
Final report, Month 36 Final report on Months 31–36	**MS6** (M34) Twelve months of live operation completed. Pass: D5.1 with the bulletin log and verified outcomes **MS7** (M36) Hand-over to the end-user. Pass: D6.2 procedures adopted; the end-user's staff use the bulletins with the project team only in support	D6.1 (M33) Software and data release D5.1 (M34) Live demonstration report D6.2 (M36) Operating procedures and policy brief D7.3 (M36) Final report

Critical path and buffers

Step on the critical path	Months	Ends at	Buffer
Network design (T4.1)	M1–6	MS1	Equipment ordered in Month 4, before the design is final, for sensors certain to be needed
Procurement and installation (T4.2)	M4–12	MS2	Installation window of 8 months; highest-value sensors installed first
Calibration of the coupled model (T2.3)	M12–18	MS3 (gate)	Model built in stages from Month 7, so calibration starts as soon as data stream
Certified surrogate (T2.4)	M15–22	MS4 (gate)	Starts before calibration ends, on preliminary parameters
Live operation (T5.2)	M23–34	MS6	Back-testing (to Month 27) runs in parallel, not before
Hand-over (T6.4)	M30–36	MS7	The end-user's staff use all bulletins from Month 34; procedures drafted from Month 30

A slip of up to two months at the go/no-go gate (MS4) is absorbed: live operation then runs from Month 25 to Month 36, still twelve months, and the end-user's staff still use the bulletins in the final months. A longer slip triggers the fallbacks of the Risk Assessment (R5, R6) and is decided by the steering committee.

What is new in this schedule

- **Every SPS report carries a decision, not only progress.** The reports fall on the evidence: Month 6 on the agreed network design (MS1), Month 18 on the calibration gate (MS3), Month 30 after the go/no-go and miss-rate gates (MS4, MS5), Month 36 on the hand-over (MS7). NATO sees each pass test and its result when it is decided.
- **Gates, not only dates.** Live operation starts only after the coupled model passes its validation test (MS3) and the surrogate meets its error bound (MS4); the certified miss rates are confirmed in back-testing (MS5). Each gate has a pass test and a fallback fixed in advance.
- **A full year live, with slack.** Back-testing runs alongside live operation, so twelve full months of operation fit even after a two-month slip at MS4, and they include every season of the hazards.
- **The hand-over is inside the schedule.** Trained staff of the Ministry of Infrastructure and Energy, the end-user, use the bulletins in Months 34–36, with the project team only in support, so the end state is demonstrated before the grant ends.